

3.6 The Chain Rule

How do we differentiate $F(x) = \sin(x^2 - 4)$? This function is the composite $f \circ g$ of two functions $y = f(u) = \sin u$ and $u = g(x) = x^2 - 4$ that we know how to differentiate. The answer, given by the Chain Rule, says that the derivative is the product of the derivatives of f and g . We develop the rule in this section.

Derivative of a Composite Function

THEOREM — The Chain Rule If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x , then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x , and

$$(f \circ g)'(x) = f'(g(x)) g'(x).$$

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$$

Where $\frac{dy}{du}$ is evaluated at $u = g(x)$.

① **Case 1** If $y = [g(x)]^r$, then $y' = r g^{r-1}(x) g'(x)$.

Example 1 Find the derivative of $f(x) = (3x^2 + 1)^2$.

Example 2 Let $y = (5x^3 - x^4)^7$, find y' .

Example 3 Let $y = \frac{1}{3x-2}$, find y' .

Example 4 Let $y = \sqrt{x^2 + 1}$, find y' .

Example 5 Let $y = \sqrt[3]{2x^2 + 3x}$, find y' .

Example 6 Let $y = \sin^5 x$, find y' .

Example 7 Show that the slope of every tangent line to the curve $y = \frac{1}{(1-2x)^3}$ is positive.

Example 8 Differentiate $h(x) = \left(\frac{x}{x+1}\right)^2$.

Example 9 Find the derivative of $f(x) = \left(\sqrt{x^2+1}+1\right)^2$.

Example 10 Differentiate $f(x) = \left(2x^3 - \sqrt{3x^4 - 2}\right)^3$.

② **Case 2** If $y = \sin g(x)$, then $y' = \cos g(x) \cdot g'(x)$.

Example 11 Let $y = \sin(x^2 + x)$, find y' .

③ **Case 3** If $y = \cos g(x)$, then $y' = -\sin g(x) \cdot g'(x)$.

Example 12 Let $y = 5\cos^{-4} 2x$, find y' .

Example 13 Differentiate

(a) $f(x) = \cos(x^2 + 1)$.

(b) $g(x) = x^2 \sin(3x) - \cos(5x)$.

(c) $h(x) = \sin \sqrt{2x^3 + 3x}$.